

Kevin Jiang

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Experience

Software Engineer Co-op, Royal Canadian Air Force (RCAF) | *Vue.js, Node.js, PostgreSQL* Jan 2026 - Apr 2026 | [About via the RCAF Digital Hub at the Department of National Defense](#)

- Built and deployed **two web applications** for RCAF operations, streamlining workflows and improving access to data
- Implemented RBAC in AWS Cognito whilst incorporating **domain-driven design**, enhancing security with modularity
- Designed and tested **50+ RESTful API endpoints** and **15+ PostgreSQL schemas** powering core app features
- Built **10+ Vue.js webpages** and components, collaborating with UI/UX designers for **user-centric design**
- Reviewed **25+ pull requests**, improving code quality, correctness, and data consistency

Data Engineering Research Assistant, University of Waterloo | *Python, R, HTML, JS* May 2025 - Current

- Aggregated **10.7 million** geospatial data points from **22 sources** into PostgreSQL databases using ArcGIS and Python
- Built and documented **5 reusable R/Python modules** for data analysis use with the **Climate Risk Research Group**
- Developed **5 web dashboards** using RMarkdown, RShiny to visualize assorted research data
- Prototyped approaches integrating novel data and techniques into **flood loss models**, evaluating impact on accuracy

Software Engineer Intern, ReliablyME | *Python, OpenAI, Express.js* Jan 2025 - Apr 2025 | [Github](#)

- Black-box tested **2400+ lines** of RESTful API code used in AI-driven commitment tracker startup, improving reliability
- **Implemented and tested features** allowing users to backdate or update their dates of commitment completion

Projects

1st Place, Factualial @ SOA Research Challenge | 68 teams | *R, Python, NumPy* Mar 2025 | [Paper](#) | [GitHub](#)

- Awarded **\$10000** for designing a national flood insurance program with **policy experts** using **layered risk modeling**
- Built **machine learning pipelines** using Google's YDF for EDA, parameter tuning, and model validation in R/Python
- Featured on **SOA podcast**, **The Actuary**, ImpACT Conference, and the **UWaterloo Math Department**

2nd Place, UTRAHacks 2025 | 249 participants | *C++, Arduino, Robotics* Jan 2025 | [GitHub](#)

- Built **Arduino-based control systems** for a robotics navigation and path-finding competition
- Programmed **C++ state machine algorithms** to complete path-finding challenges

3rd Place, Debug Debacle @ McHacks 2025 | 471 participants | *Python, React.js, SocketIO* Jan 2025 | [GitHub](#) | [Link](#)

- Published **full-stack multiplayer game** with OAuth and Vercel where players race to **fix AI-generated bugged code**
- Deployed **real-time validation framework** to ensure players correctly solved GumloopAI AI-generated bugs
- Implemented front-end with **React.js**, back-end with **Flask**, and **SocketIO** for network communication

Firmware Member, Waterloo | *C++, STM32, CAN bus protocol* Sep 2024 - Present | [Github](#)

- Integrated new **Raspberry Pi as central control** unit of the design team's machine via CAN protocol
- Developed and documented **STM32 scripts** using **C++** to verify proper operation of the CAN network

Software Member, Midnight Sun | *Python, AWS, SQL, Conda, Linux, Bash* Sept 2024 - Present | [Github](#)

- Developed **PostgreSQL database** for compute race route efficiency calculations using **Conda and Python**
- Wrote **protocols and documentation** for remote management of computed route calculations using Linux and Bash

Education

University of Waterloo, Bachelor of Software Engineering Sep 2024 – Apr 2029

Select scholarships and awards:

1st of 75, \$2000 prize - CPA Business Case Challenge

Top 20 Finalist - Canadian Physics CAP Exam(Olympiad)

National Recipient - Power and Glider Pilot Scholarship

Gold Award - Canadian Chemistry Contest

Skills and Qualifications

- **Languages:** C, C++, JavaScript, Typescript, Python, R, SQL, HTML/CSS, Bash, Java, Scala
- **Libraries/Frameworks:** Vue.js, React.js, Axios, Express.js, Pinia, SQLite, STM32, CASBIN, Geopandas, Pandas, Shiny, Tkinter, Flask, NumPy, Tensorflow, PyTorch, RMarkdown, Bootstrap, Datatables, Quill.js, Vuetify, Vueflow, Leaflet
- **Tools/Technologies:** Node.js, Docker, Google Cloud Platform, AWS, MSSQL Server, PostgreSQL, MongoDB, Sequelize, Apache, Nginx, Office 365, Windows, Debian, Ubuntu, Raspbian, AutoCAD, Hazus, ARCGIS, QGIS